Google Cloud H2S

AI Builder Cup

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# Team OpsCrew

## Expense exception plans with human confirmation

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Theme: Future of Work & Enterprise Productivity

Precloud review draft / 11 October 2026

# Brief about the idea

OpsCrew helps operations staff turn a synthetic expense exception into a supported handling plan.

Gemini extracts facts, proposes steps, then reviews the proposal in a separate call. Fixed policy checks constrain each stage.

A person reviews the current plan and explicitly confirms a case. The app reads the saved record independently before showing success.

The prototype covers one synthetic USD policy. It creates handling records and performs no reimbursement or payment.

# Opportunities and proposed value

Problem: staff must reconcile intake text, missing receipts and policy thresholds before deciding what to request next.

Proposed difference: OpsCrew binds the plan to supported facts and exact policy clauses, then stores the person's confirmation with that version.

Design value: input edits invalidate old plans. Contradictions and unsupported citations block case creation.

Expected impact: clearer exception triage and fewer unsupported actions. Actual staff time savings and customer adoption remain unmeasured.

# Features offered by the solution

Synthetic presets and original text citations make intake decisions inspectable.

Facts, Planner and Reviewer outputs expose disagreements and exact policy evidence.

Firebase visitor identity, owned history and version-bound confirmation protect each person's handling records.

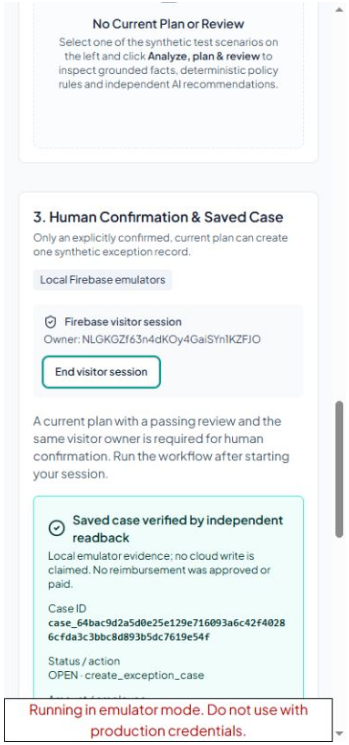
Atomic writes, idempotent recovery and independent readback prevent duplicate or unverified success. Shared quotas bound model traffic.

# Process flow and confirmation




Missing evidence produces a request plan. Contradictions block progression.
Editing intake requires a new analysis and a new confirmation.

# Interface and narrow-screen review




## Observed local browser behavior

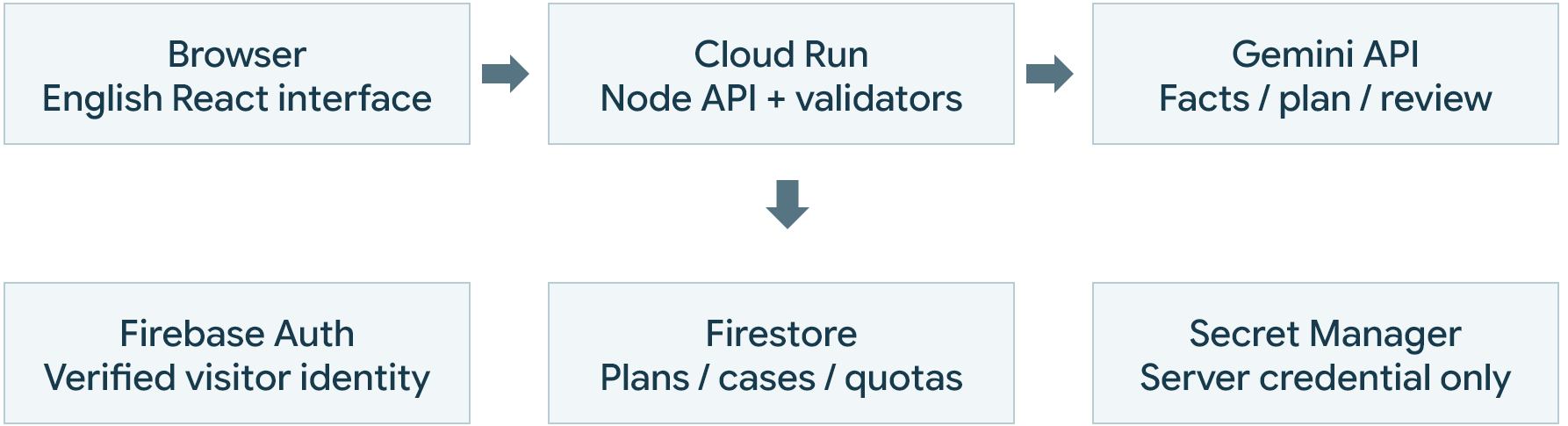
Explicit confirmation remains separate from the AI review.

Saved history survives a page reload with the same visitor identity.

The 390 px viewport had no horizontal overflow. Keyboard focus was visible.

Screenshot: local Auth/Firestore emulators and injected model output, 10 October. Public cloud behavior awaits acceptance.

# Architecture of the proposed cloud deployment




The server verifies ownership and mediates every business write. Browser Firestore access is denied. Cloud deployment remains the next team gate.

# Technologies used in the solution

| Component | Role |
| --- | --- |
| Google AI Studio Build | Fresh application origin on 7 October |
| Gemini 3.1 Flash-Lite | Facts, separately prompted Planner and Reviewer |
| Node 24 / React / TypeScript | API, explicit review interface and fixed validators |
| Firebase Auth + Firestore | Visitor ownership, atomic records and shared quotas |
| Cloud Run + Secret Manager | Planned public runtime and server-only credential |

Local extensions are disclosed. No code from the earlier non-Google prototype entered this repository.

# Implementation cost and traffic controls

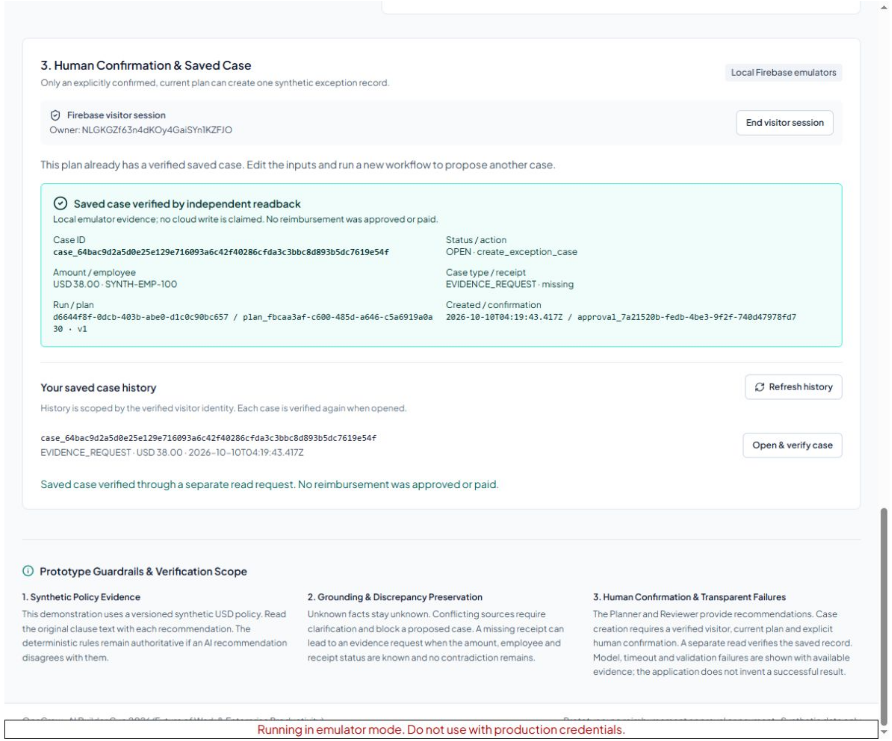
Team management target: USD 40. Billing alerts and application quotas cannot guarantee a monetary spending cap.

Default model limits: 2 concurrent runs globally, 1 per visitor. Each visitor has 2 runs per minute and 12 per UTC day.

Global limits: 6 runs per minute and 60 per UTC day. Each workflow reserves up to 6 provider attempts. Unknown usage receives no refund.

Record tokens, elapsed time and failures from real runs. Historical live failures have unknown usage and billed cost. Verify actual cloud pricing before activation.

# Snapshots of the prototype




## Synthetic USD 38 missing-receipt example

The plan requests a receipt.

A person confirms the handling record. The API then independently reads the saved case.

No payment occurs.

Local emulators + injected models.
10 October 2026

# Performance report and evidence boundaries

| Evidence | Result | Method |
|---|---|---|
| Offline engineering | 216 / 216 | Policy / HTTP / UI regressions |
| Auth + Firestore | 27 / 27 | Real local emulators, injected models |
| Public semantic cases | 12 / 12 | Injected models, MemoryCaseStore |
| High-risk repeats | 12 / 12 | 4 predefined public cases, each 3 runs |
| Fault / recovery | 6 / 6 | Injected failures and lost responses |
| Production packaging | 9 HTTP + 12 artifact groups | Pruned native runtime, no Docker daemon |

Current local Gemini: 3 public branches passed after proxy repair. Three earlier failures remain recorded. Cloud stability and independent held-out acceptance remain pending.

# Additional details and future development

Next acceptance: deploy the frozen release to Cloud Run with real Firebase identity, Firestore and a server-only Gemini secret.

The teammate seals eight independent cases before first evaluation. Retain every attempt and run the main demo successfully three times in sequence.

Then record the real cloud flow, publish the product and sub-three-minute English video, and replace the draft proposal's pending link statuses.

After evaluation, consider broader policies and staff timing studies. Production reimbursement integrations require a separate design and authorization.

# Repository, demo and final product links

## Public GitHub repository

github.com/sundaylee3100-ljl/AI-Builder-Cup-2026-OpsCrew

Public demo video: pending final cloud recording
Final product URL: pending teammate cloud deployment

This proposal is a precloud review draft. The team must add working public links and verify anonymous access before contest submission.